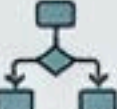




Smart Parking with Vision AI: Project Documentation & Strategic Overview

Comprehensive Analysis, Methodology, and Implementation Plan



 Agenda	 Current Situation	 Project Overview
 Success Criteria	 Project Plan	 Tech Stack
 Schedule & Budget	 Risk Mitigation	 Unforeseen Challenges

Current Situation: The Parking Challenge in Ontario & Toronto

Economic Impact



1.4 BILLION

Annual Congestion Value Loss (Toronto)

Verified 2023 Stats: Economic drain due to parking-related congestion.

Environmental Cost



95 L
Fuel Wasted per Vehicle/Year



220 kg
CO₂ Emitted per Vehicle/Year

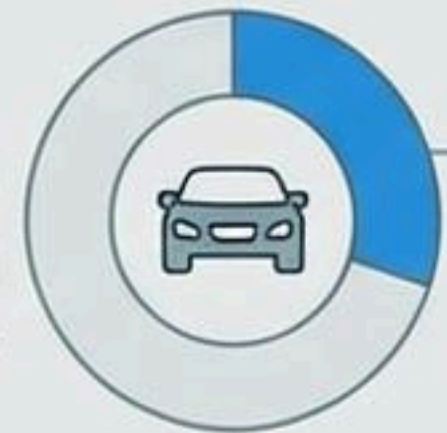
Significant environmental footprint from circling vehicles.

Urban Efficiency Drain



5.4 MINUTES
Average Search Time (Ontario Curb Parking)

30%
of Downtown Traffic



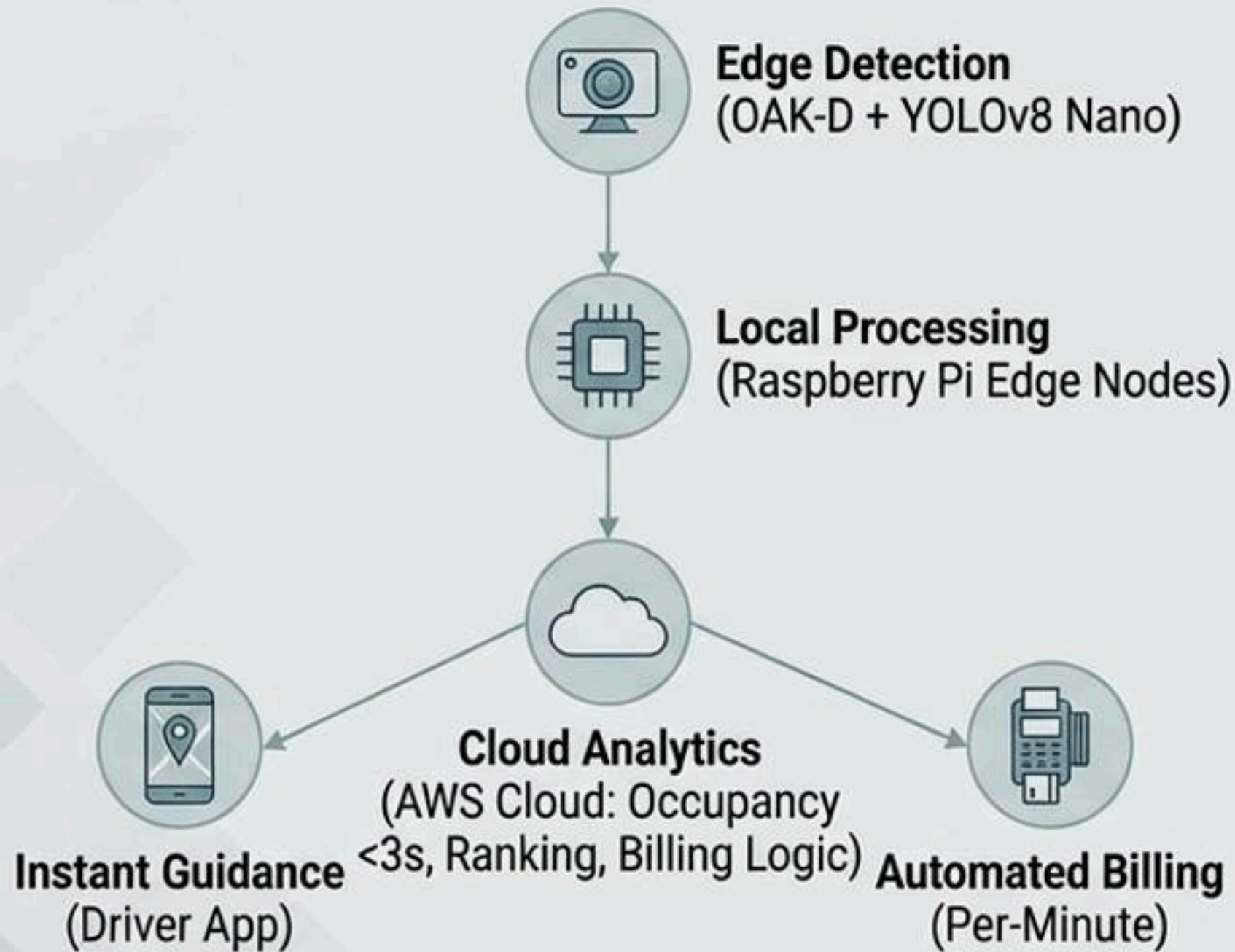
Circling for Spots

Critical drain on urban mobility and time.

These **verified 2023** stats reveal economic, environmental, and urban efficiency drains that demand an AI-driven, real-time remedy.

Project Overview: AI-Powered Smart Parking System

Integrated Tech Stack & Workflow



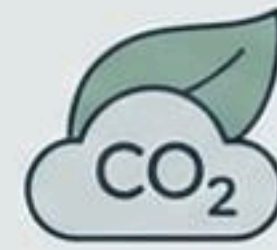
Real-time detection & processing from edge to cloud.

Key Project Outcomes for Ontario



Reduced Search Time

Directs drivers to nearest free spot, eliminating circling.



Lowered Emissions

Decreases vehicle idling and unnecessary fuel consumption.



Minimized Congestion

Improves overall traffic flow in and around lots.

Transforming parking efficiency and sustainability across Ontario.

Success Criteria: Measurable Thresholds for Acceptance



Real-Time Accuracy:
>85%

High detection confidence.



Spot Ranking:
Correctly Ranked

Optimal spot identification.



End-to-End
Latency: <3 s

Sustained fast response.



Simultaneous
Spaces: 20+

Handles concurrent volume.



Live Data:
Shown on Frontend

Real-time user display.



Billing & Demo:
To the Minute, Dec 8
Demo Passed

Precise charging & successful milestone.

These measurable thresholds define project completion and de-risk stakeholder acceptance.

Project Plan

5-Week Scrum Cadence: Methodology, Development, and Integration Roadmap

Scrum Cadence & Governance



Daily:
Stand-ups
(Quick Check-ins)



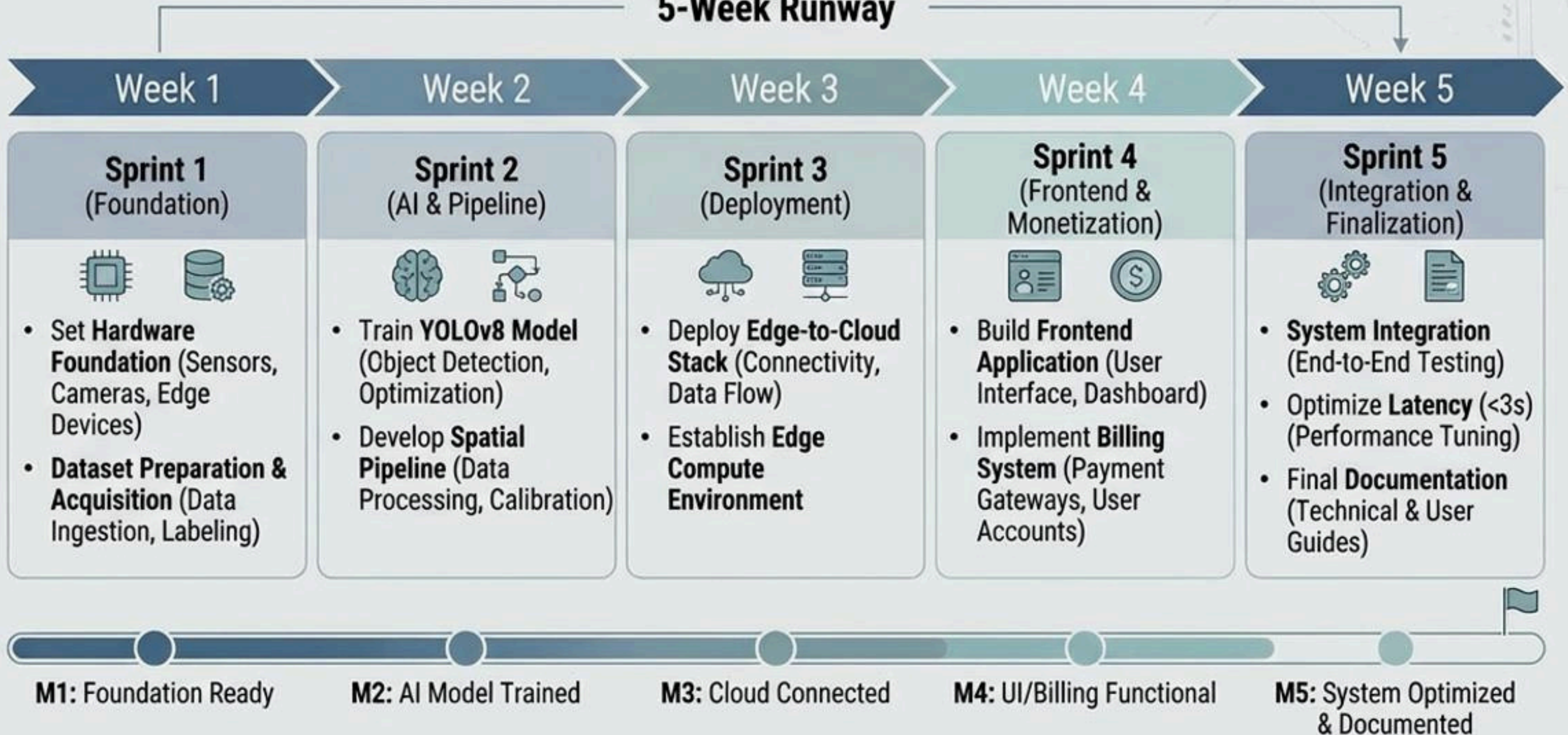
Weekly:
Sunday Reviews
(Milestone Assessment)



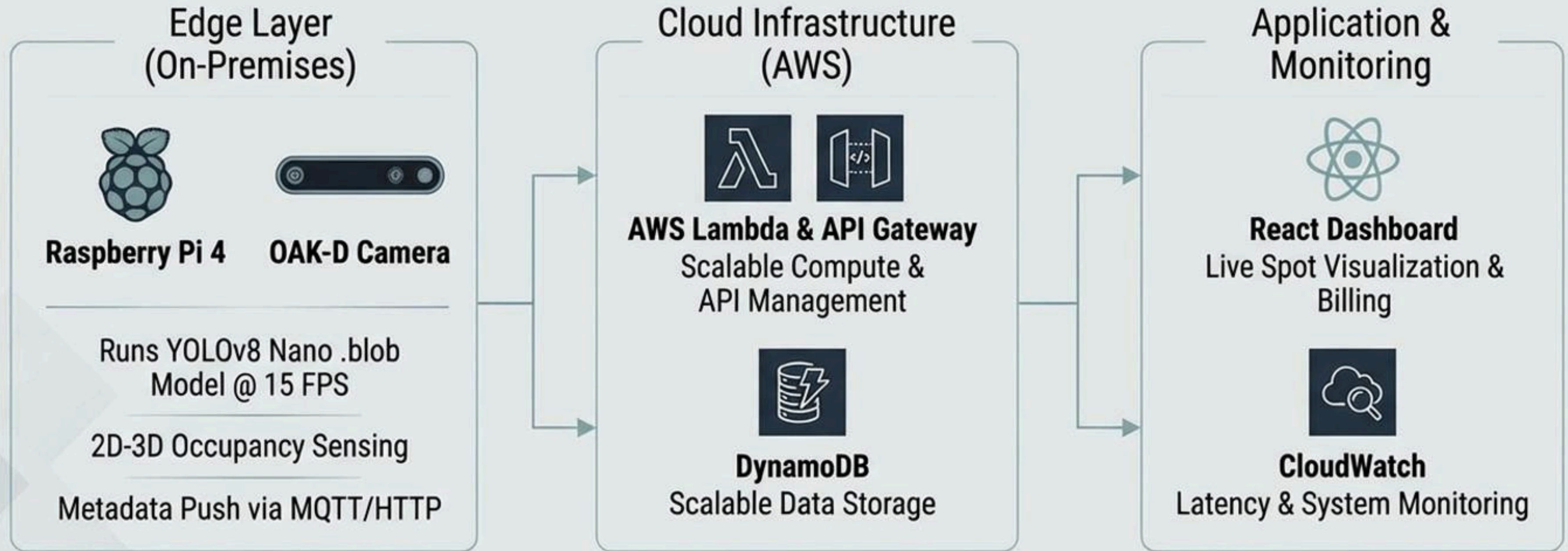
Process:
Definitions of Done
(Clear Acceptance Criteria)

Key Milestones & Deliverables

5-Week Runway



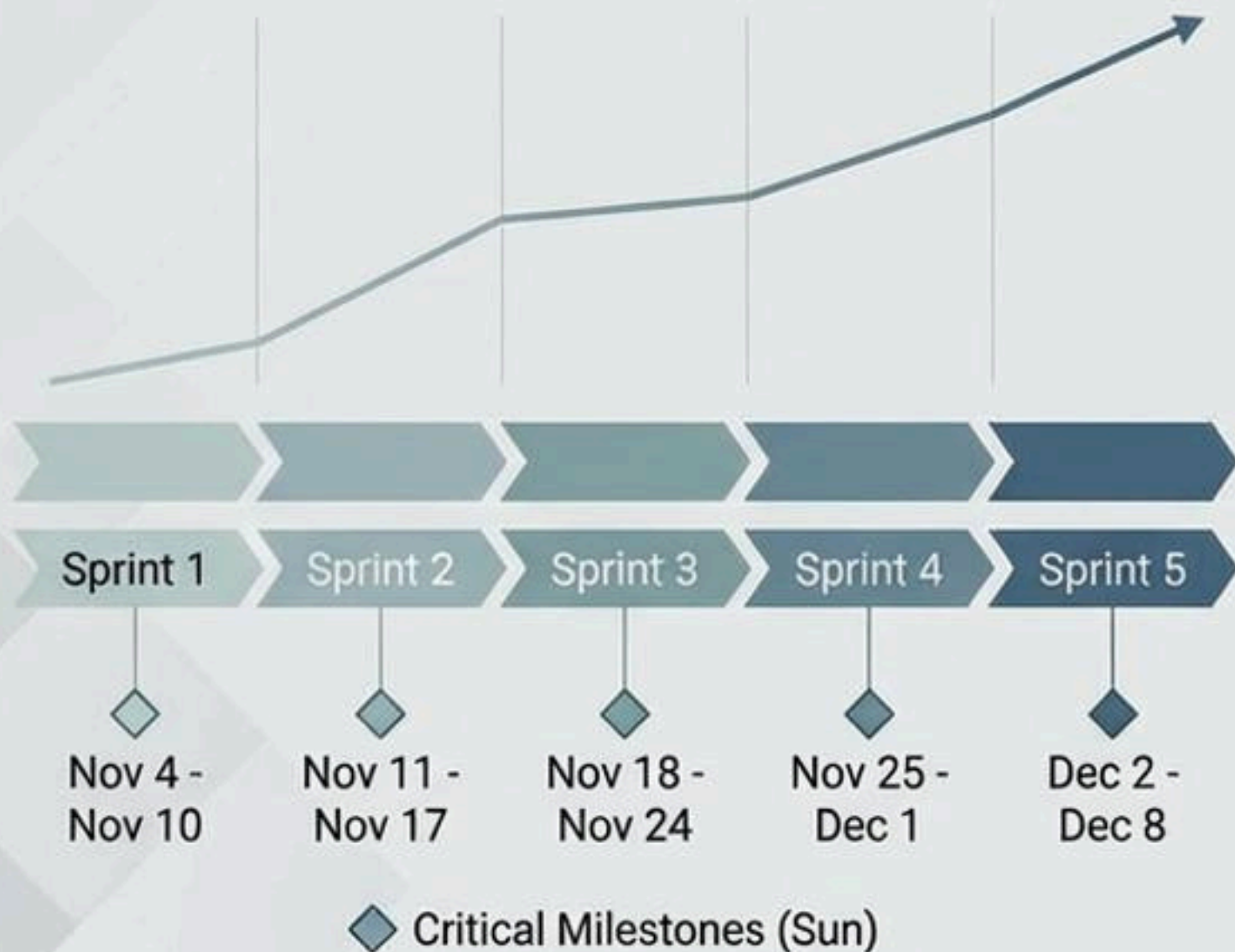
Tech Stack: Edge, Cloud & AI Integration



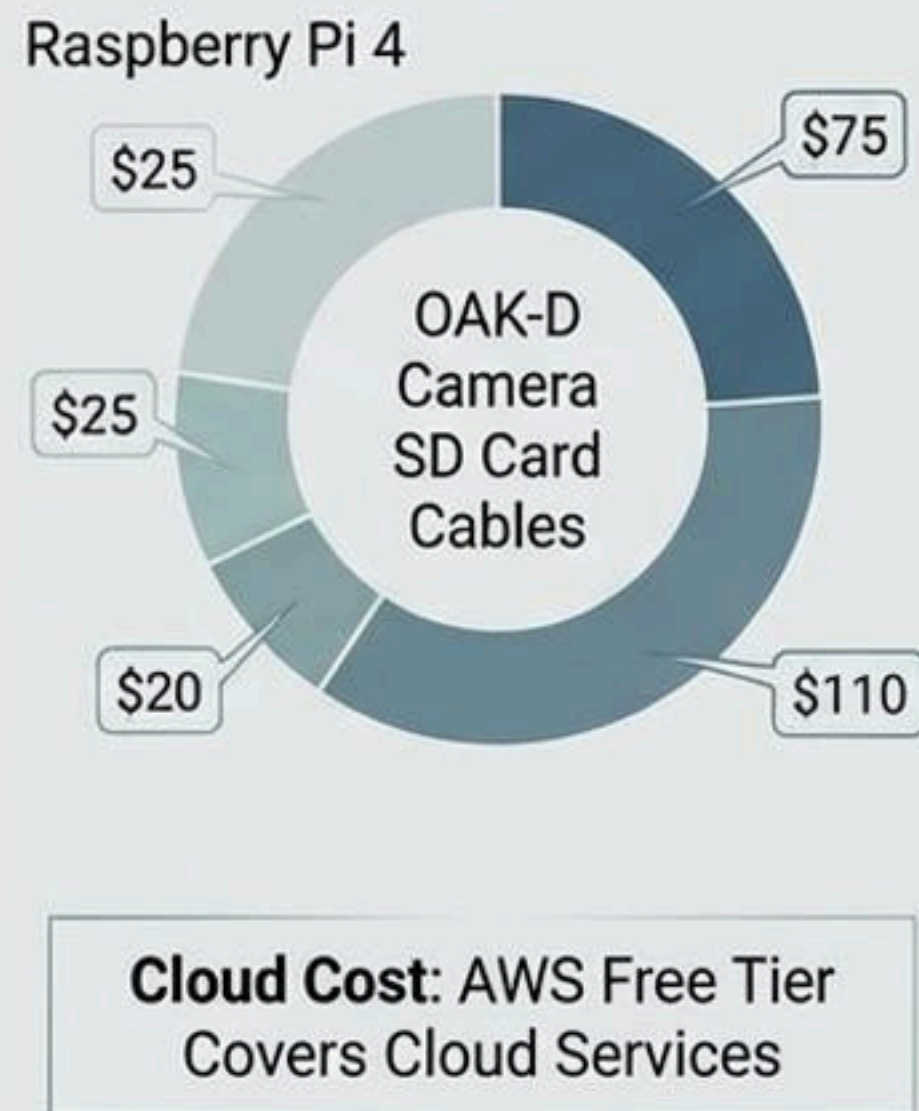
Vision, Edge, and Cloud Act as One Seamless System

Project Schedule, Budget & Resources Overview

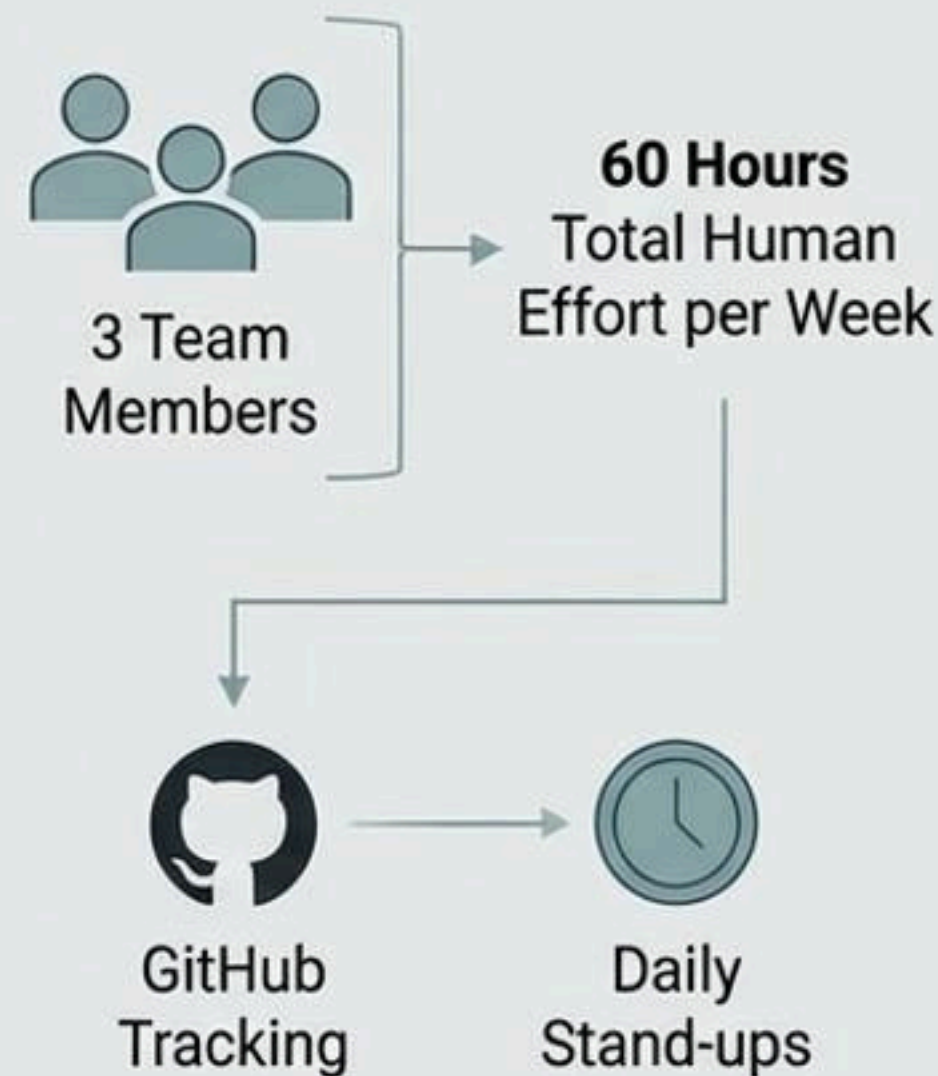
Project Schedule (5 One-Week Sprints)



Budget Overview (\$230 Hardware Spend)



Resource Allocation & Tracking (60 hrs/week)



Project Risk & Mitigation Strategies

Risk Item & Trigger	Mitigation & Response Strategy
1. Latency (>3s)	 FPS Throttling & Client-Side Prediction
2. Model Accuracy (<85%)	 Data Augmentation & Pre-Trained Weights
3. Hardware Failure	 Webcam Fallback System
4. Integration Issues	 Weekly Tests & Interface Contracts
5. AWS Cost Overrun	 Billing Alerts (\$5, \$10, \$15) & Nightly Shutdowns

Unforeseen Challenges & Project Strategies

Challenges & Scope Risks



Network Outages & Depth-Camera Drift at Demo



Sudden Scope Creep & Feature Requests



Potential Delays to Dec 8 Deadline

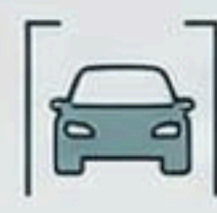
Mitigation & Response Strategies



Sprint 5 Buffer Tasks & Pre-recorded Backup Demos/Offline Datasets



Strict Change-Control: Team Approval for >2 Hrs Work



Future-Work Parking Lot for Enhancements (Post-Deadline)

